Movie Sentiment Analysis: Neural Network Performance Report

1. Student Information

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Framework: PyTorch

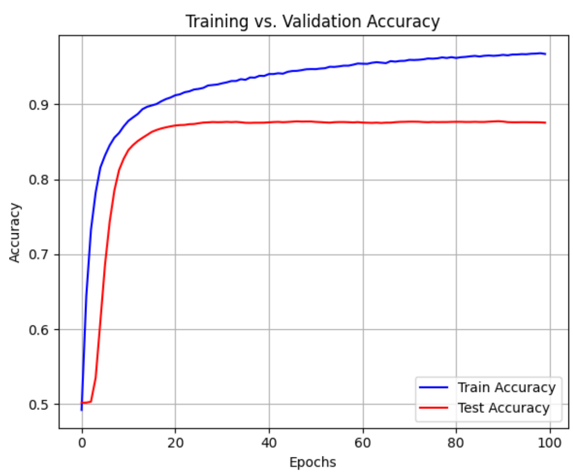
2. Experiment 1: Baseline Architecture

Configuration: 5,000 Features | 32 Hidden Neurons | Learning Rate: 0.001 | Dropout: 0.5

Epoch	Train Loss	Test Loss	Train Acc	Test Acc
10	0.4097	0.6806	87.08%	82.77%
50	0.2532	0.5085	94.72%	87.72%
100	0.1763	0.3071	96.74%	87.57%

Insight (Exp 1):

Experiment 1 showed rapid learning, but the gap between Train Accuracy (96.7%) and Test Accuracy (87.5%) indicates mild overfitting. The model began memorizing training noise in later epochs.



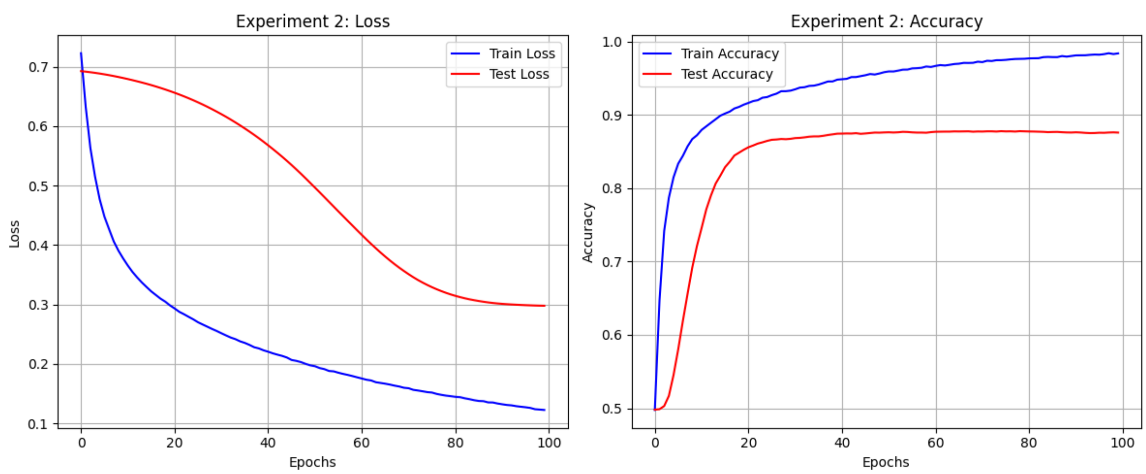
3. Experiment 2: Optimized Architecture

Configuration: 5,000 Features | 64 Hidden Neurons | Batch Normalization | LR: 0.0007

Epoch	Train Loss	Test Loss	Test Acc
10	0.3775	0.6809	72.13%
50	0.1977	0.5043	87.62%
100	0.1227	0.2979	87.60%

Insight (Exp 2):

By increasing capacity (64 neurons) and reducing the Learning Rate to 0.0007, the model reached a much lower Test Loss (0.2979). This reduction in loss indicates higher confidence and better generalization.



4. Final Comparison & Verdict

Metric	Experiment 1	Experiment 2
Hidden Neurons	32	64
Learning Rate	0.001	0.0007
Test Accuracy	87.57%	87.60%
Final Test Loss	0.3071	0.2979

Conclusion:

Experiment 2 is the superior model. It provides a more stable loss curve and better error reduction. The use of Batch Normalization helped in stabilizing the 64-neuron architecture.